# Optimal Incentive Design for Decentralized Dynamic Matching Markets

**Chen Chen**
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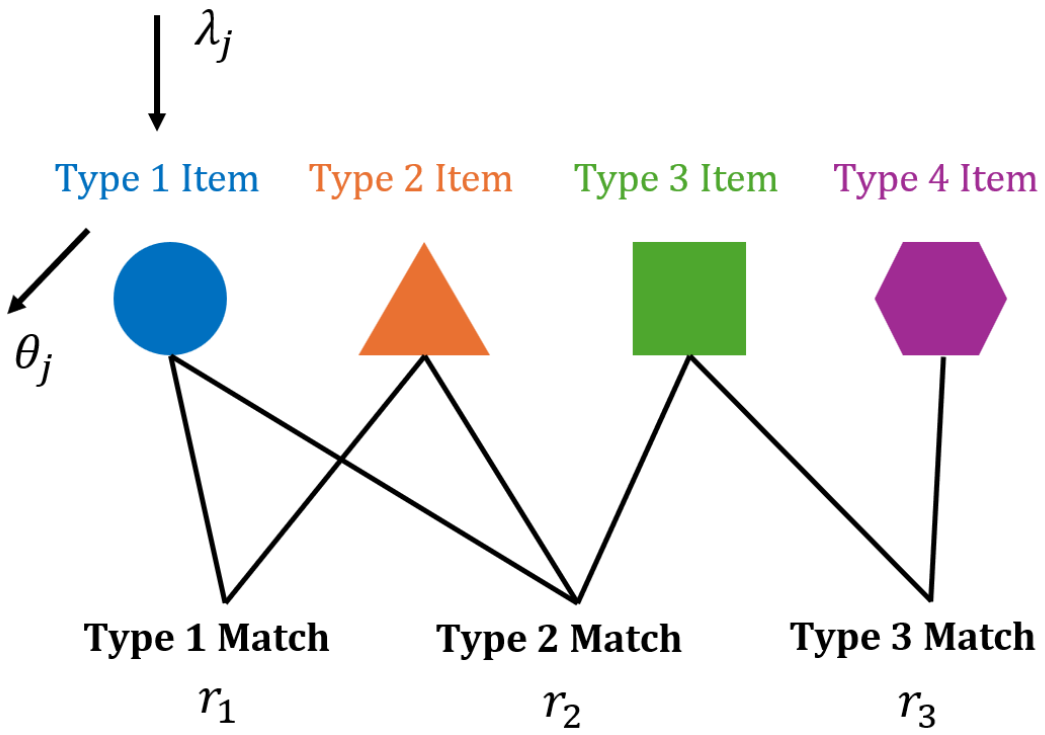
March 2026

# Dynamic matching




- Heterogeneous item and match types
- Stochastic arrivals and departures with rates $\lambda_j$ and $\theta_j$

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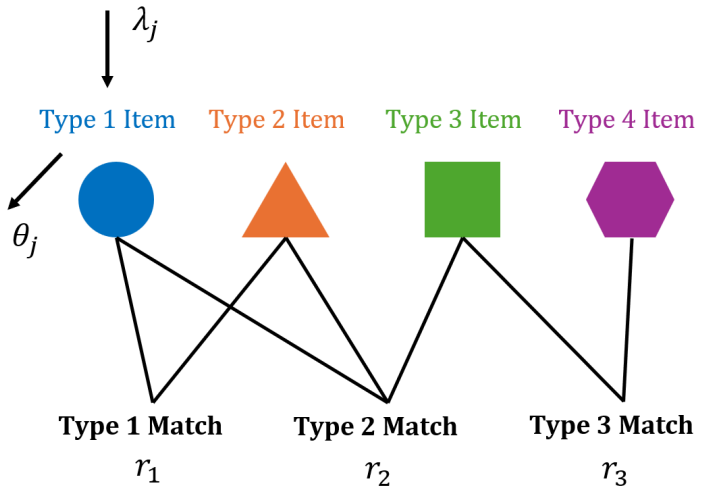
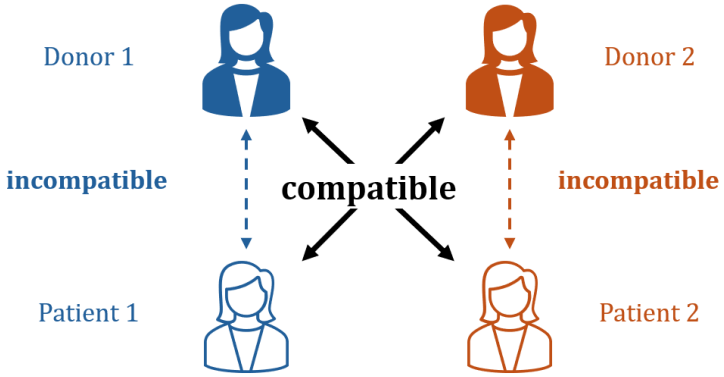



Examples: kidney exchange



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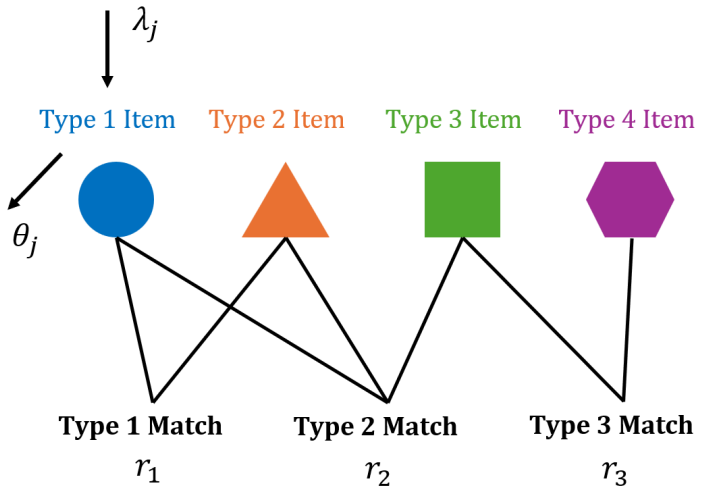
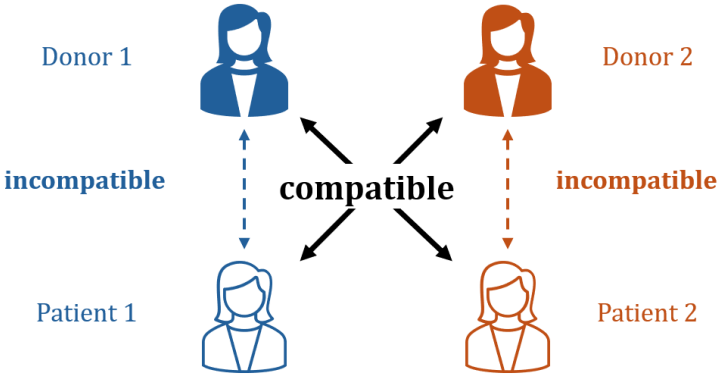
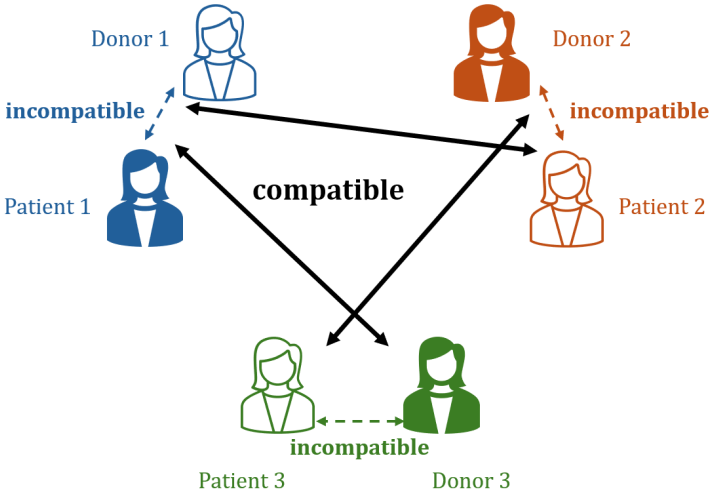



Examples: kidney exchange (multi-way matches possible)



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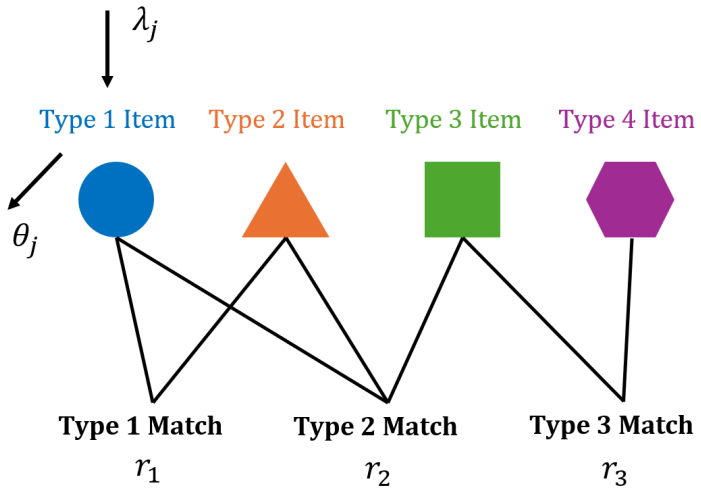


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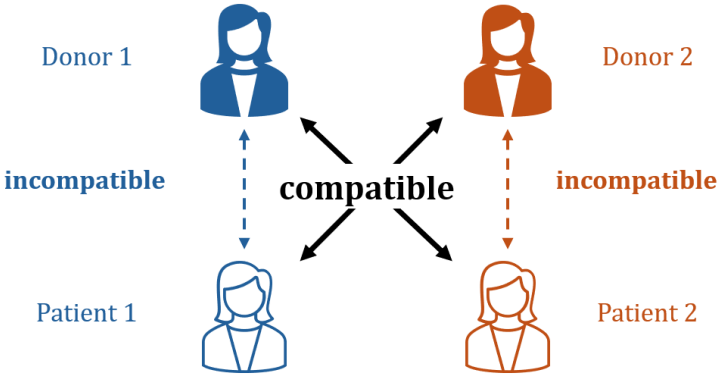
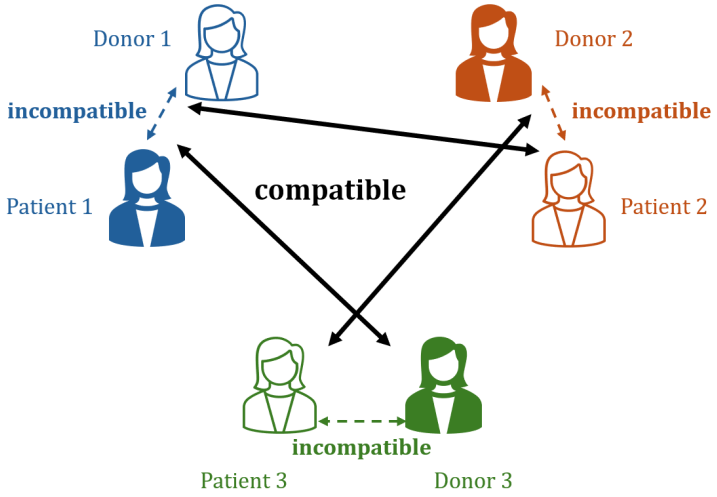


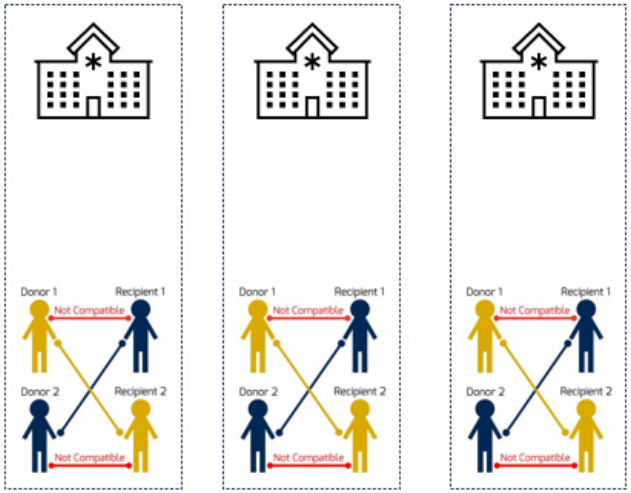




- **Decisions:** when and how to match
- **Objective:** maximize long-run average match values

# Decentralized dynamic matching

We focus on *decentralized* setting:

- Example: multi-hospital kidney exchange

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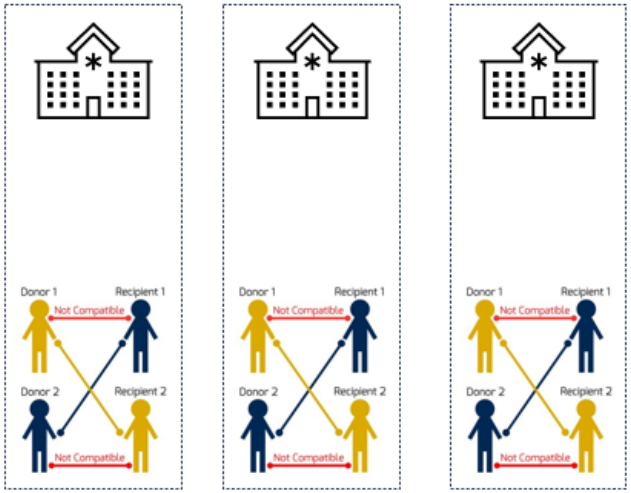


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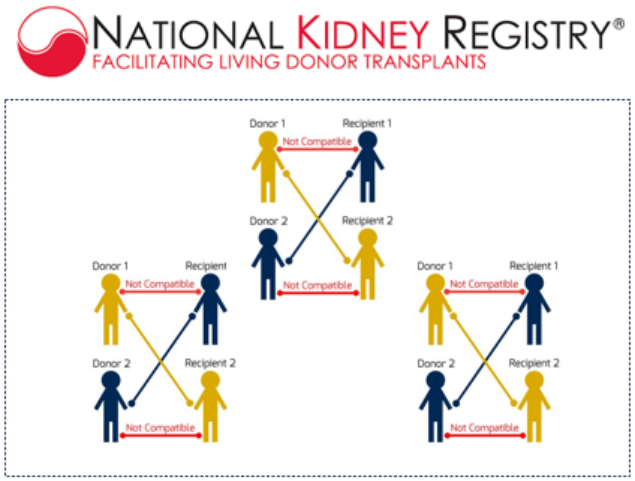


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**thicker market = better matches**

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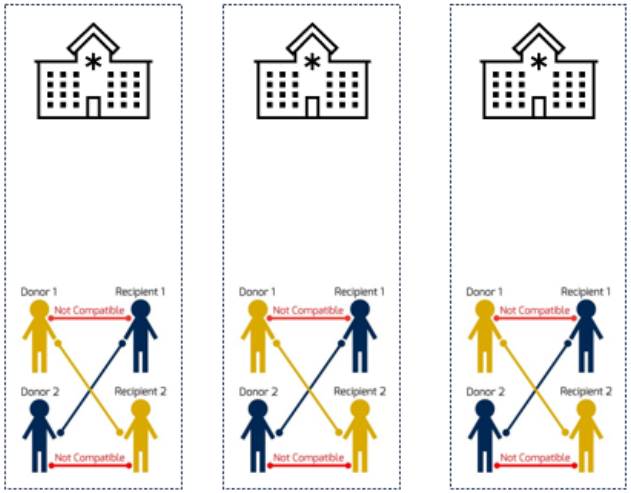


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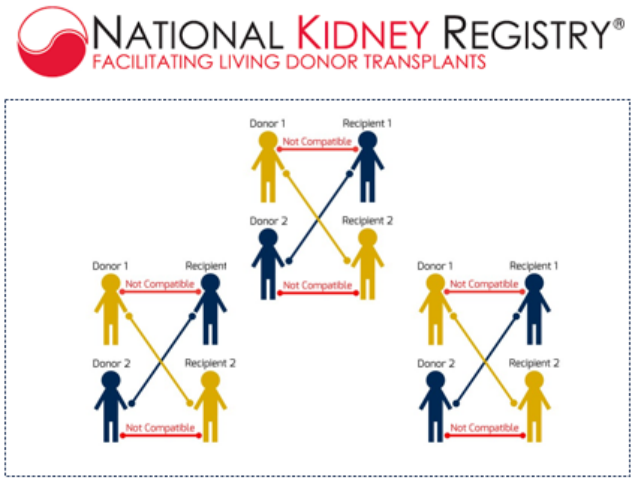


- Agents (e.g., hospitals) each manages its own stream of items
- **Goal:** mechanism to incentivize full participation $\Rightarrow$ maximizes social efficiency
- **Challenges** beyond algorithm design: handling strategic behaviors

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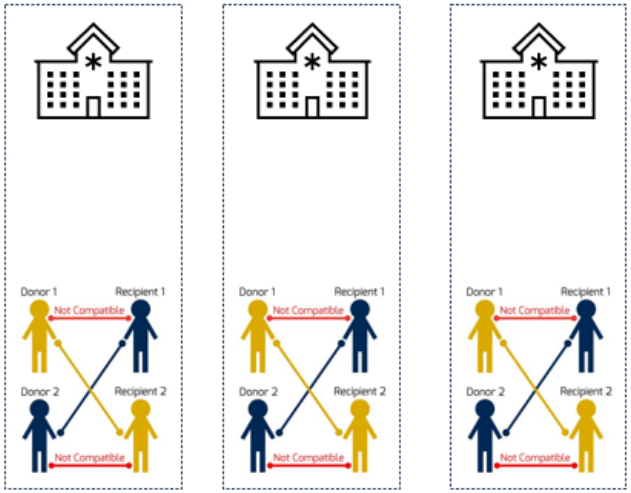


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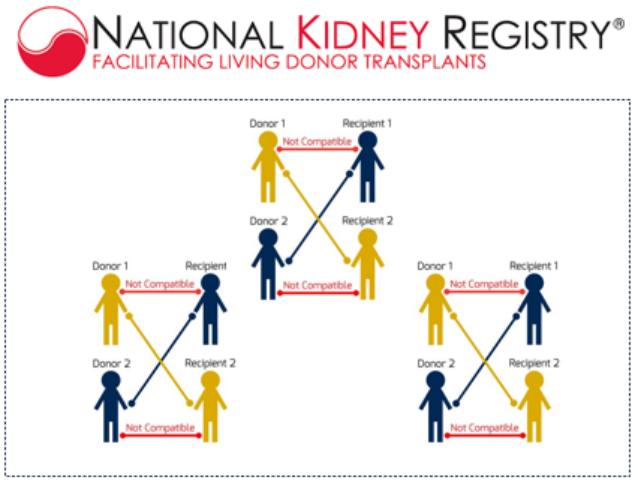


- **Main result**: develop simple and intuitive monetary and non-monetary mechanisms that incentivize complete item submissions when # agents is large.

$\Longrightarrow$ System dynamics and performance match those under centralized control in large markets

# Related Literature

**Dynamic matching in centralized settings (no incentive constraints):**

- Aouad and Sarıtaç (2022); Aveklouris et al. (2025); Akbarpour et al. (2020); Ashlagi et al. (2023); and many others.
- Strategic behavior of market participants is abstracted away.

**Mechanism design in static settings (no dynamics):**

- Ashlagi and Roth (2014); Toulis and Parkes (2015); Ashlagi et al. (2015); Agarwal et al. (2019).
- Do not account for real-world frictions such as limited lifespans and stochastic arrivals and departures.
- Mechanisms may become infeasible or inefficient in dynamic settings.

A marketplace design problem:

Incentivize resource pooling, in private information setting

for queueing, **matching**, and general stochastic systems

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Chen, Chen, Qian (2025)
Incentivizing Resource Pooling.
*Management Science*.

# Model setup

- $N$ strategic agents: Each agent $i$ manages a local dynamic matching
  - Type-$j$ items arrive at rate $\lambda_{ij}$ and depart at rate $\theta_j$
  - Performing type-$m$ match generates reward $r_m$
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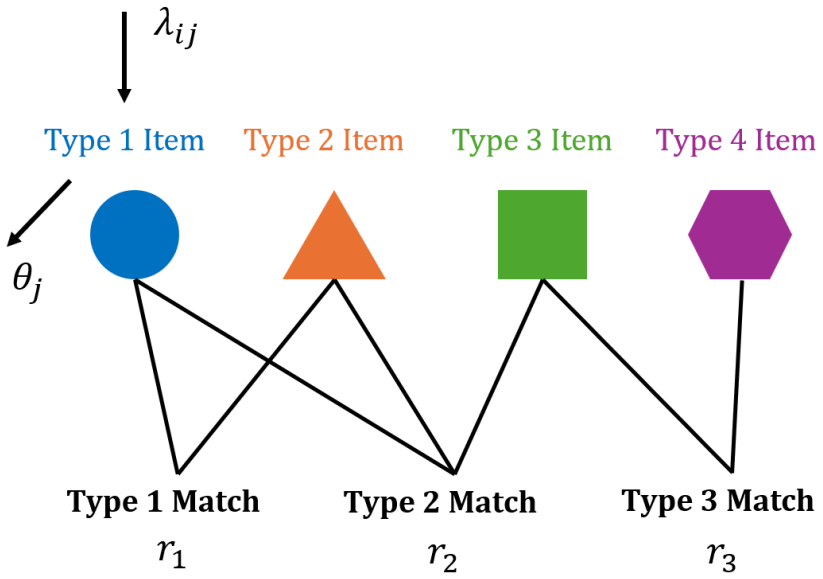

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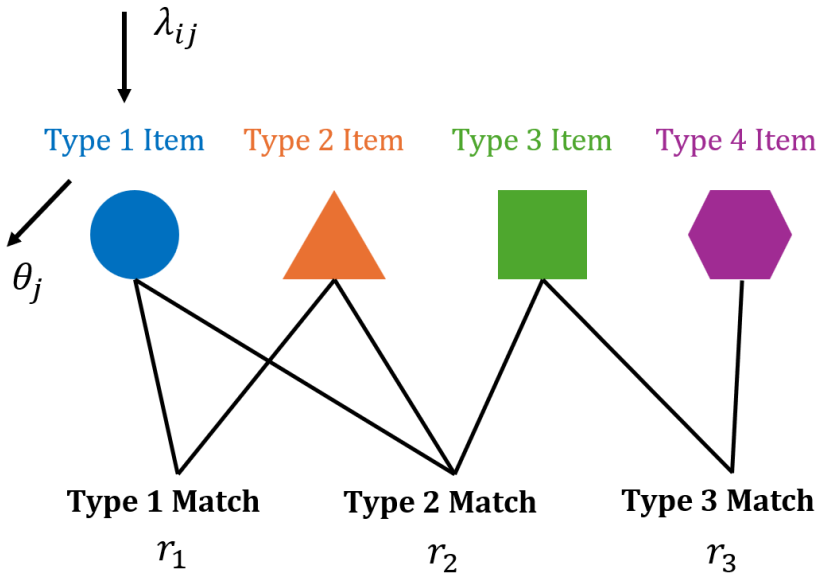
- Private information: item arrivals and actions <u>not observable</u> by others



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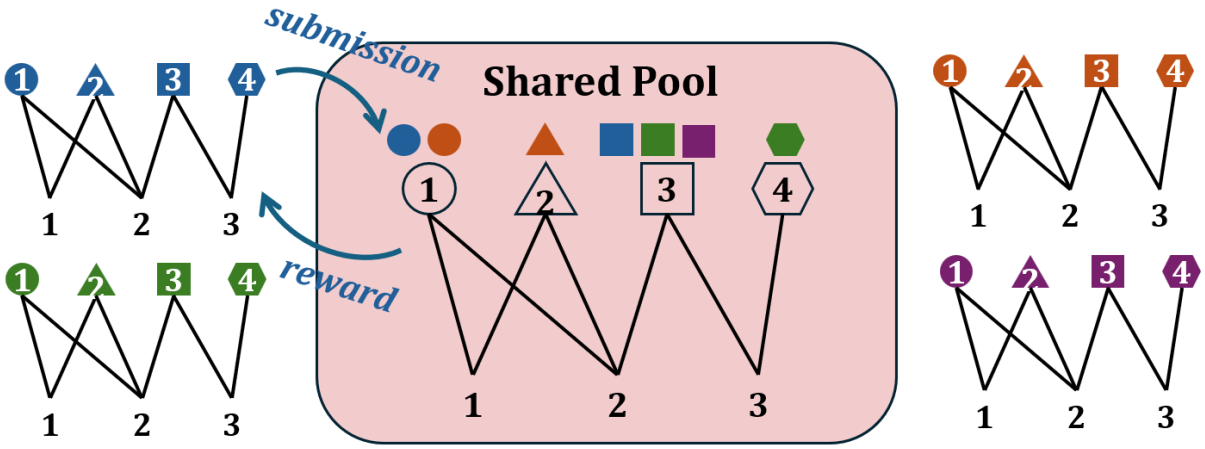

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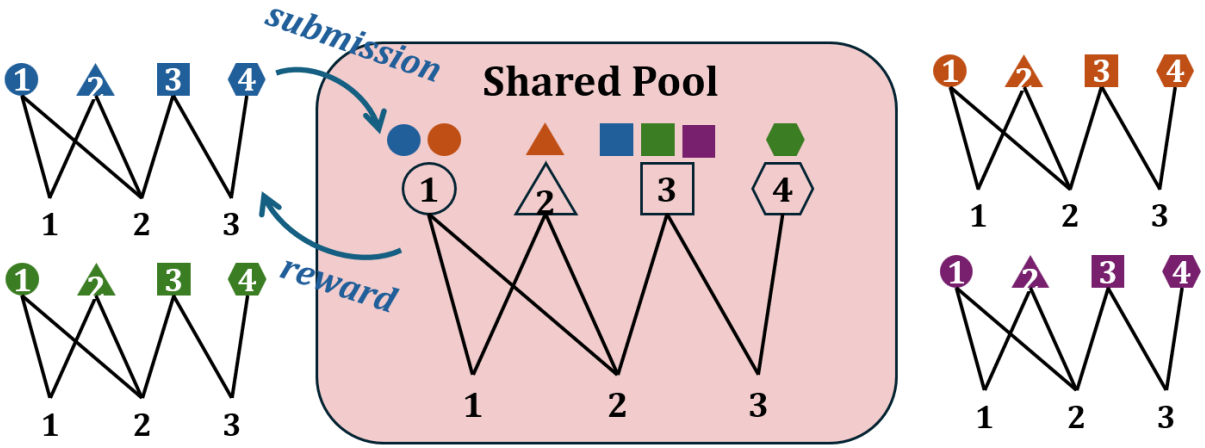

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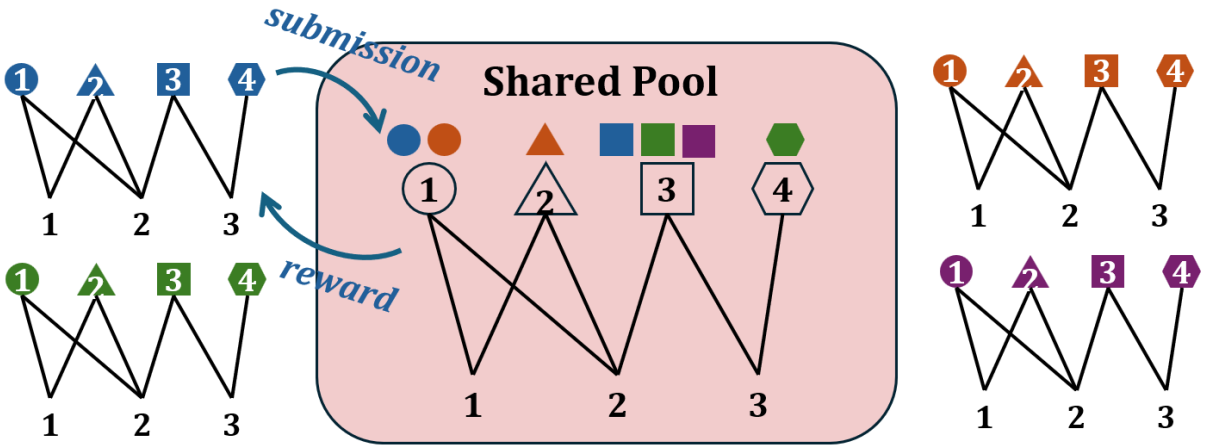

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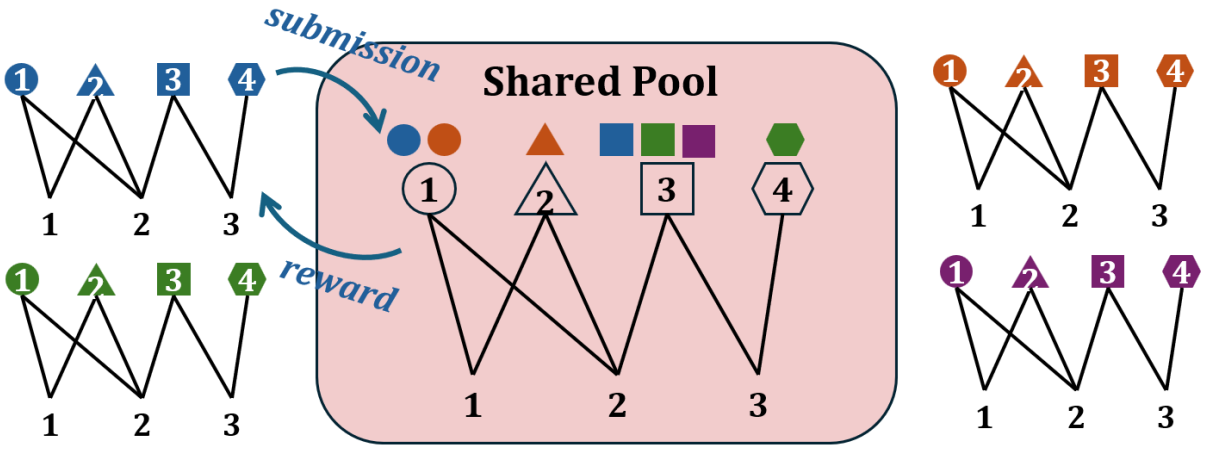
- **Objective of designer:** maximize total match values of the system in the long-run average
- Huge design space: reward can depend on item type, matching outcome, and can be monetary or non-monetary (e.g., priority, credits)



# Benchmark: centralized control

Fluid relaxation of centralized problem:

$$\max_{x_m \geq 0} \quad \sum_{m=1}^{K} r_m x_m$$

$$\text{s.t.} \quad \sum_{m=1}^{K} M_{jm} x_m \leq \lambda_j, \, \forall \, j \leq J.$$

$\lambda_j = \sum_{i=1}^{N} \lambda_{ij}$: aggregate item arrival rate

**(capacity constr.)**

number of type-$j$ items in match $m$

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$x_m^*$: **matching rate** of match $m$

- $x_m^* > 0$: $m$ is essential match
- $x_m^* = 0$: $m$ is non-essential match

**Dual:**

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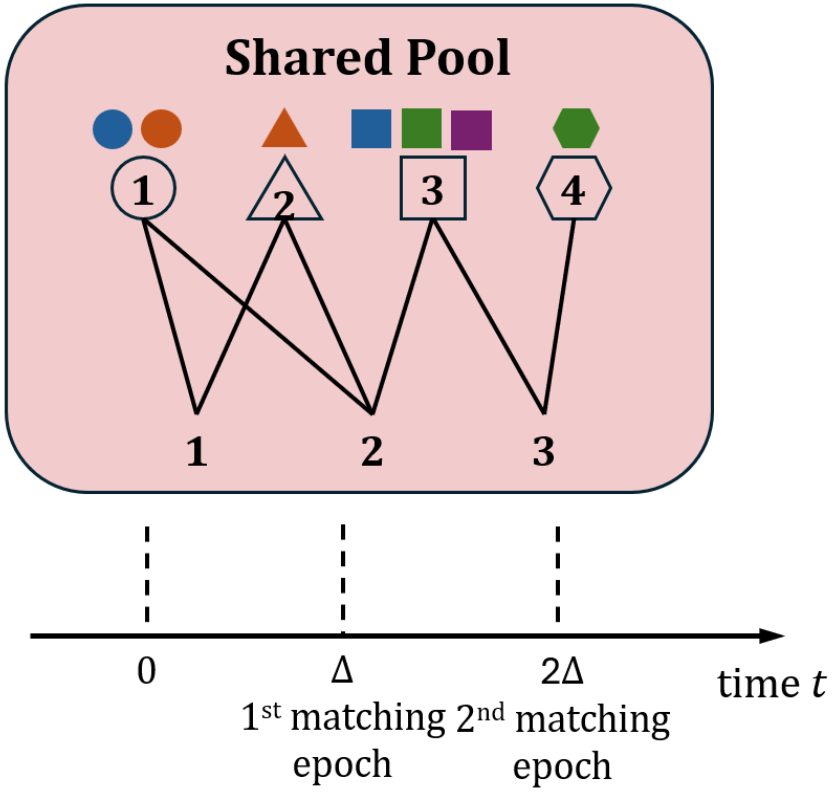
- Centralized matching at shared pool: Any algorithm that is (i) asymptotically optimal as $N \to \infty$ and (ii) restricted to $\mathcal{M}_+$

# Centralized matching at shared pool

Example: Periodic matching policy

- Shared pool performs matching every $\Delta = o(1)$ (e.g., $\Delta = N^{-1/3}$).
- Performance of our mechanism depends on regret of centralized matching.



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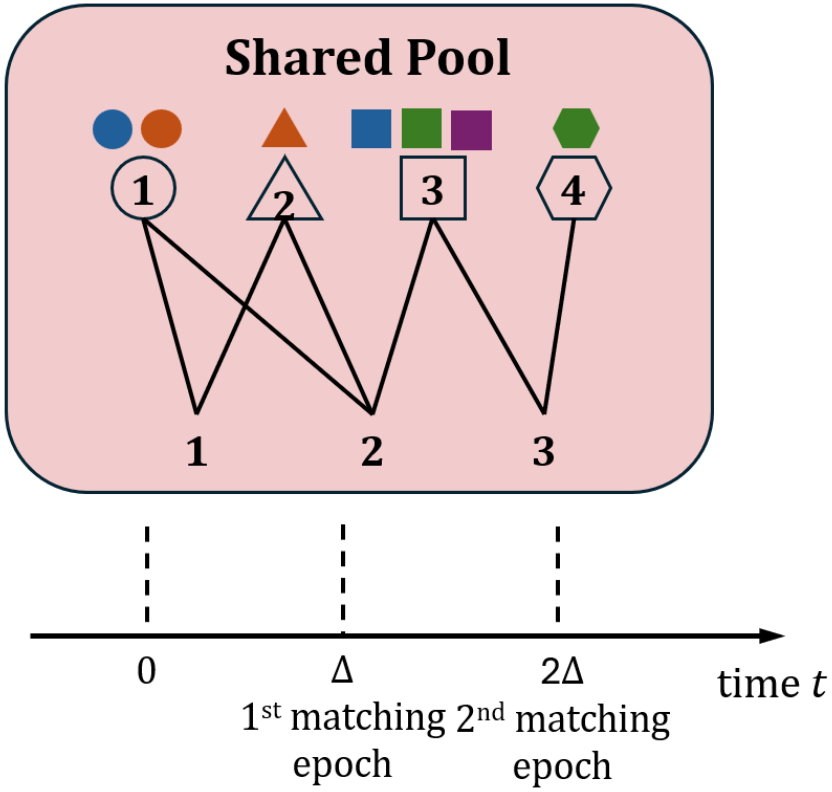

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- Theoretical guarantees:
  - Full submission is approximate Nash equilibrium under MV
  - Full submission guarantees a stronger mean-field equilibrium under MVC

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$\Longrightarrow$ For *any* agent, full submission is optimal

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**Theorem.** Under the MV mechanism, full item submission by all agents constitutes an approximate Nash equilibrium.

*Intuition:* over-demanded jobs get matched with probability one as $N \to \infty$, which ensures an expected payoff of $p_j^*$.

# Marginal-Value (MV) mechanism

Let us examine the incentive structure:

- Favorable incentives for over-demanded items: when matched with high probability in shared pool (i.e., when $N$ is large)
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- Favorable incentives for over-demanded items: when matched with high probability in shared pool (i.e., when $N$ is large)
  - Holding is strictly suboptimal
  - Is either submitted or internally matched immediately upon arrival
- Incentive issues for under-demanded items: since $p_j^* = 0$,
  - No incentive to submit
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$$\Longrightarrow \text{the MVC mechanism!}$$

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A stronger equilibrium concept than approximate Nash equilibrium

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  $\implies$ reduces to a Markov decision problem
- **Mean-field equilibrium (MFE):**

$$\boxed{\boldsymbol{w} = H(\boldsymbol{w})}$$

conjectured matching probs

induced matching probs when all agents follow best responses to $\boldsymbol{w}$

# Marginal-Value-plus-Credit (MVC) mechanism

- **Observation:** submissions of <u>under-demanded items</u> need to be rewarded, even though their marginal value is zero.
- Simply tweaking the allocation probs in MV mechanism does *not* make full submission an MFE.

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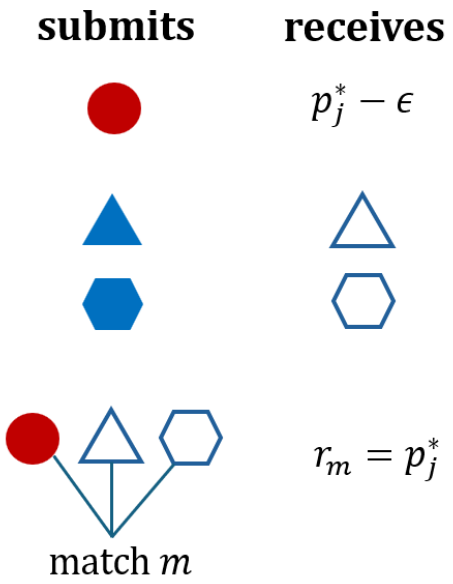


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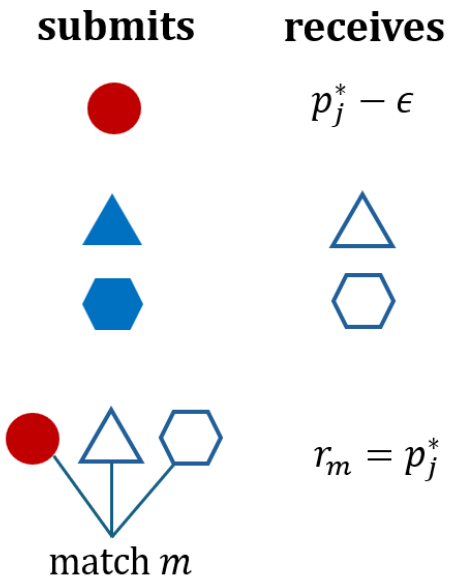


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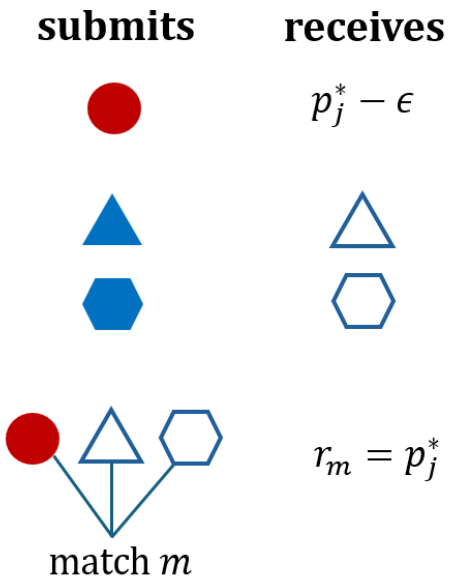
- Expired credits convert into one-time lotteries for collected taxes $\epsilon$.
- Mechanism can be implemented without money via randomized matching allocations.

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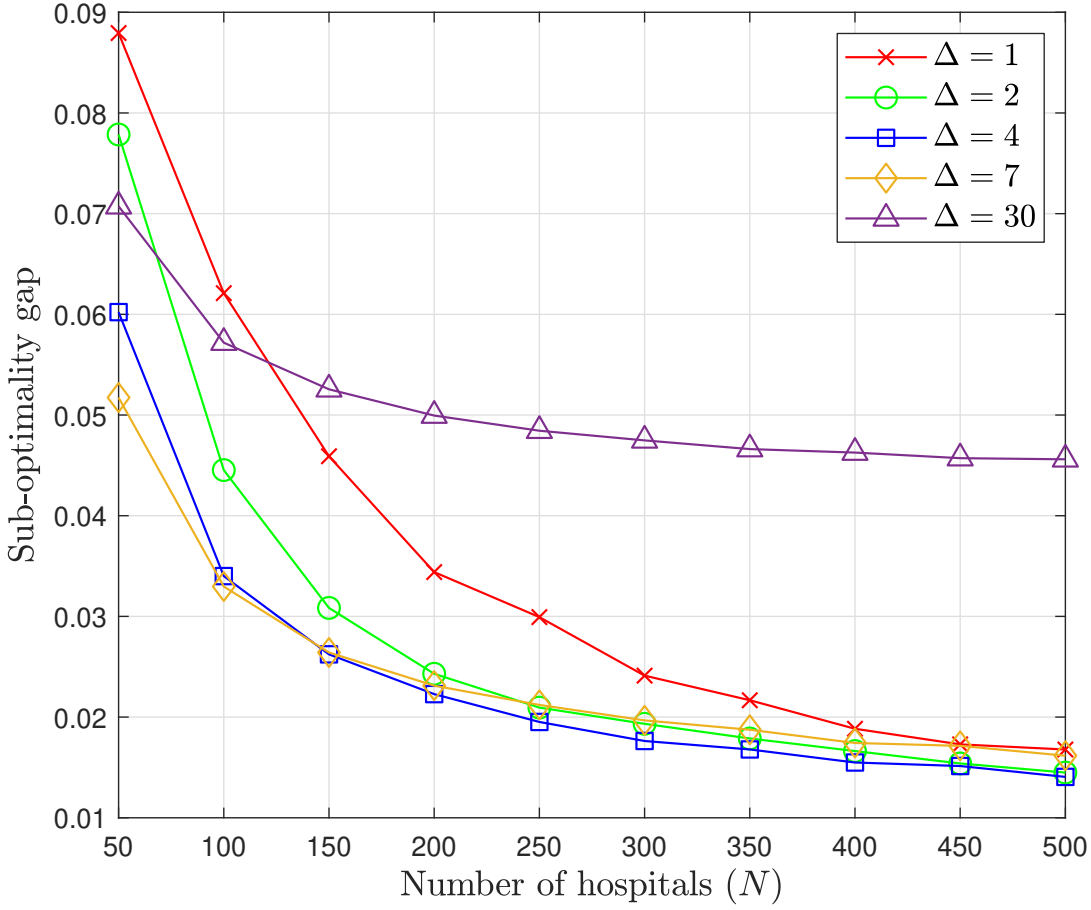
# Marginal-Value-plus-Credit (MVC) mechanism

**Theorem.** Under the MVC mechanism, full item submission by all agents constitutes an MFE when $N \geq N_0$ (some constant) and also an approximate Nash equilibrium.

# Numerical results

Multi-hospital kidney exchange example based on real data

$$\text{Sub-optimality gap} = \frac{\text{Fluid relaxation bound} - \text{Payoff from full submission}}{\text{Payoff from full submission}}$$



# Summary

- We study optimal incentive design in decentralized dynamic matching, where agents have limited information about others (so deviation cannot be punished directly)

- **Operational takeaway:** simple marginal-value based mechanisms incentivize **full** item submission in large markets
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**Reference:**

C. Chen, P. Qian, and J. Zhang. 2026. Optimal Incentive Design for Decentralized Dynamic Matching Markets. Minor Revision at *MS*. **(Matching)**

C. Chen, Y. Chen, and P. Qian. 2025. Incentivizing Resource Pooling. Forthcoming at *MS*. **(Queueing)**

# Appendix